

Extension and Exhumation of the Hellenic Forearc Ridge in Kythera

A. Marsellos, W. Kidd • Published in *The Journal of geology* 23 May 2008 • Geology

Detailed mapping and new structural observations on Kythera demonstrate that there is a major detachment fault, here termed the Potamos detachment fault (PDF), exposed on the northern part of the island and bordering the domed structure of a metamorphic core complex. The detachment forms the contact of contrasting metamorphic and nonmetamorphic units, and both ductile and brittle structures are found, related to its extensional development. The earliest Potamos mylonites, as well as some other...

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E. Skourtsos S. Lekkas Geology • 2011

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Quaternary E-W Extension Uplifts Kythira Island and Segments the Hellenic Arc

Gino de Gelder D. Fernández-Blanco +5 authors R. Lacassin Geology • Tectonics • 2022

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Exhumation of HP-rocks accompanied by low-angle normal faulting and associated detachment fault of Milos Island - Evidence from zircon fission-track thermochronology

A. Marsellos J. Garver K. Kyriakopoulos Geology · 2012

Southward roll-back and slab retreat of the subducting African plate has been the cause of extension and magmatism in the Aegean since the Early Miocene. The Hellenic subduction system has shown a... [Expand](#)

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Formation and fragmentation of a late Miocene supradetachment basin in central Crete: implications for exhumation mechanisms of high-pressure rocks in the Aegean forearc

W. Zachariasse D. V. van Hinsbergen A. Fortuin Geology · 2011

We present a new lithostratigraphy and chronology for the Miocene on central Crete, in the Aegean forearc. Continuous sedimentation started at ~10.8 Ma in the E–W trending fluvio-lacustrine Viannos... [Expand](#)

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Variations in fault-slip data and cooling history reveal corridor of heterogeneous backarc extension in the eastern Aegean Sea region

U. Ring K. Gessner S. Thomson Geology, Environmental Science · 2017

Abstract We report fault-slip data across the boundary between the highly extended and largely submerged crust underlying the Aegean Sea from Samos in the north to eastern Crete in the south, and the... [Expand](#)

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Map view restoration of Aegean–West Anatolian accretion and extension since the Eocene

D. V. Hinsbergen S. Schmid Geology · 2012

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Normal Faulting Along the Kythira-Antikythira Strait, Southwest Hellenic Forearc, Greece

V. Veliz-Borel V. Mouslopoulou A. Nicol J. Begg O. Oncken Geology, Environmental Science ·

[Frontiers in Earth Science](#) · 2021

Upper-plate normal faults along forearcs often accumulate slip during >Mw 6 earthquakes. Such normal faults traverse the forearc of the Hellenic Subduction System (HSS) in Greece and are the focus of... [Expand](#)

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Cenozoic tectonic evolution of Asia: A preliminary synthesis

A. Yin Geology · 2010

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Vertical-axis rotations accommodated along the Mid-Cycladic lineament on Paros Island in the extensional heart of the Aegean orocline (Greece)

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Rapid Exhumation of High-Pressure Metamorphic Rocks in Kythera-Peloponnese (Greece) Revealed by Apatite (U-Th)/He Thermochronology

A. Marsellos K. Min D. Foster Geology · [The Journal of geology](#) · 2014

Apatite (U-Th)/He and apatite fission-track cooling ages from high-pressure (HP) metamorphic rocks in the South Aegean forearc along the Kythera Strait (Kythera and southeastern Peloponnese) suggest... [Expand](#)

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Structure and kinematics of Upper Cenozoic extensional detachment on Naxos and Paros (Cyclades Islands, Greece)

P. Gautier J. Brun L. Jolivet Geology · 1993

This paper presents a study of the ductile and brittle deformation on Naxos and Paros islands (Cyclades, Greece). Previous maps and studies of the two islands have shown that a major low-angle fault... [Expand](#)

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Kinematics of the central Hellenides

T. Doutsos G. Pe-Piper K. Boronkay I. Koukouvelas Geology · 1993

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The late Alpine evolution of an extensional shear zone, Naxos, Greece

I. Buick Geology · [Journal of the Geological Society](#) · 1991

The mid-Miocene structural evolution of the metamorphic complex on Naxos involved deformation in a kilometre-scale low angle ductile shear zone. The structures described are consistent with... [Expand](#)

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Exhumation Processes : Normal Faulting , Ductile Flow , and Erosion

Geology · 2003

Over the past 25 years, there has been a growing appreciation of the role that tectonic processes play in exhuming metamorphic rocks. This trend began with the discovery of highly attenuated crustal... [Expand](#)

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Miocene high-pressure metamorphic rocks of Crete, Greece: rapid exhumation by buoyant escape

S. Thomson

B. Stöckhert

M. Brix

Geology · [Geological Society Special Publication](#) · 1999

Abstract The pre-Neogene thrust sheets of Crete, Greece, accreted during Oligocene and Early Miocene time, can be divided into two main groups juxtaposed by a Miocene extensional detachment.... [Expand](#)

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Active deformation of a segment of arc: the strait of Kythira, Hellenic arc, Greece

N. Lyberis

J. Angelier

É. Barrier

S. Lallemand

Geology · 1982

Abstract A field analysis of Late Cenozoic faults has been carried out in the southwest Hellenic arc, on the islands of Kythira and Antikythira as well as on both sides of the strait, in southeast... [Expand](#)

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Miocene detachment in Crete and exhumation P-T-t paths of high-pressure metamorphic rocks

L. Jolivet

B. Goffé

P. Monié

C. Truffert-Luxey

M. Patriat

M. Bonneau

Geology · 1996

A major problem posed by the geology of Crete is the horizontal contact of an upper unit without Miocene metamorphism onto a metamorphosed lower one with Early Miocene high pressure/low temperature... [Expand](#)

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Regional implications of geochemistry and style of emplacement of Miocene I-type diorite and granite, Delos, Cyclades, Greece

G. Pe-Piper

D. Piper

D. Matarangas

Geology · 2002

Abstract The Miocene plutons of the Cyclades were emplaced in a subduction setting during regional back-arc extension of continental crust, that led to flat-lying mid-crustal detachment faulting.... [Expand](#)

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Detachment faults in the Aegean core complex of Ios, Cyclades, Greece

M. Forster

Gordon Lister

Geology · [Geological Society Special Publication](#) · 1999

Abstract Several generations of detachment fault systems have been recognized in the Aegean metamorphic core complex of Ios, Cyclades, Greece. Multiple strands occur in each fault system, with... [Expand](#)

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Late Cenozoic exhumation of the Cascadia accretionary wedge in the Olympic Mountains, northwest Washington State

M. Brandon

M. Roden-Tice

J. Garver

Geology · 1998

The apatite fission-track method is used to determine the exhumation history of the Olympic subduction complex, an uplifted part of the modern Cascadia accretionary wedge. Fission-track ages are... [Expand](#)

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